

Modules over PIDs

R : PID $(\mathbb{Z}, k[x], \mathbb{Q}[x])$

Defⁿ ① An R -module M is
finitely generated if there
is a finite subset $X \subseteq M$

$$\text{s.t. } \left\{ \sum_i r_i m_i : r_i \in R, m_i \in X \right\} \\ = M$$

Example:

① $M = \{0\}$, ② $M = R$
 $= \{r \cdot 1 : r \in R\}$

③ M_1, M_2 : f.s. R -modules
then $M_1 \times M_2$: f.s. R -module

$$(4) \underbrace{R \times \dots \times R}_{n\text{-times}} = R^n$$

||

$$\{r_1 e_1 + r_2 e_2 + \dots + r_n e_n : r_i \in R\}$$

where $e_i = (0, \dots, \underset{\substack{\uparrow \\ i\text{-th position}}}{1}, 0, \dots, 0)$

Defⁿ (2) $N \leq M$ is an R -submodule if N is a subgroup of M and $\forall r \in R, n \in N, r \cdot n \in N$

(3) $N \leq M$: There is a unique R -module structure on M/N s.t. π is an R -linear map

$\downarrow \pi$
 M/N

$$r \cdot (m + N) = rm + N$$

(5) $N \leq M$ is f.g. R -module



M/N is a f.g. R -module.

(6) $N \leq R$: R -submodule



$N \leq R$: ideal

Rmk: There exist commutative rings R with ideals that are not finitely generated

But when R is a PID every ideal is generated by one element! So this is not a problem.

Thm (Fundamental theorem
for f.s. modules over
PIDs)

M : f.s. R -module

(1) (Invariant factors)

$\exists$ unique integer $\overset{\text{rank}}{m} \geq 0$

and ideals $R \supseteq (f_1) \supseteq (f_2) \supseteq \dots \supseteq (f_r) \supseteq (0)$
s.t. $\underbrace{\hspace{10em}}_{\text{invariant factors}}$

$M \underset{\substack{\uparrow \\ R\text{-linear} \\ \text{isomorphism}}}{\simeq} R/(f_1) \times R/(f_2) \times \dots \times R/(f_r) \times R^m$

(2) (Elementary divisors)

$\exists$ unique integer $m \geq 0$,

irreducible elements $q_1, \dots, q_n \in R$

& integers $n_1, \dots, n_n \geq 1$ ← elem. divisors

s.t. the sequence $(q_1^{n_1}, q_2^{n_2}, \dots, q_n^{n_n})$
of ideals is unique up to permutation

s.t.

$$M \cong R / (q_1^{n_1}) \times \dots \times R / (q_n^{n_n}) \times R^m$$

Kmk (c) $R \cong \mathbb{Z}^m$: This is just
a restatement of the FT for
f.g. abelian groups

(1) $R = k[x]$: This gives the classification of finite dim'l k -pairs & hence of conjugacy classes of matrices in $M_n(k)$

Example:

$$R = \mathbb{F}_3[x]$$

$$M = \frac{\mathbb{F}_3[x]}{(x^4+1)} \times \frac{\mathbb{F}_3[x]}{(x^2+1)}$$

$x^4+1 \in \mathbb{F}_3[x]$ is irreducible.

x^4+1 : A zero for this is an element of order 8 and such an element exists in $\mathbb{F}_9 = \frac{\mathbb{F}_3[x]}{(x^2+1)}$

In fact,

$$x^4 + 1 = (x^2 + x - 1)(x^2 - x - 1)$$

$$= \frac{\mathbb{F}_3[x]}{(x^2 - x - 1)}$$

$$M = \frac{\mathbb{F}_3[x]}{(x^2 + x - 1)(x^2 - x - 1)} \times \frac{\mathbb{F}_3[x]}{(x^2 + 1)}$$

$$\cong \frac{\mathbb{F}_3[x]}{(x^4 + 1)(x^4 + 1)} \cong \frac{\mathbb{F}_3[x]}{(x^2 + x - 1)} \times \frac{\mathbb{F}_3[x]}{(x^2 - x - 1)}$$

invariant factor

Generalization
of Chinese
Remainder Theorem

$\times \frac{\mathbb{F}_3[x]}{(x^4 + 1)}$
elementary
divisors